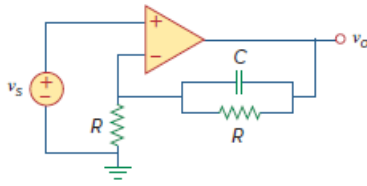


Problem

Obtain the closed-loop gain and phase shift for the circuit in Fig. 10.34. Let $R = 10 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$, and $\omega = 1000 \text{ rad/s}$.

Figure 10.34:



Step-by-step solution

Step 1 of 4

We first convert the circuit to the frequency domain

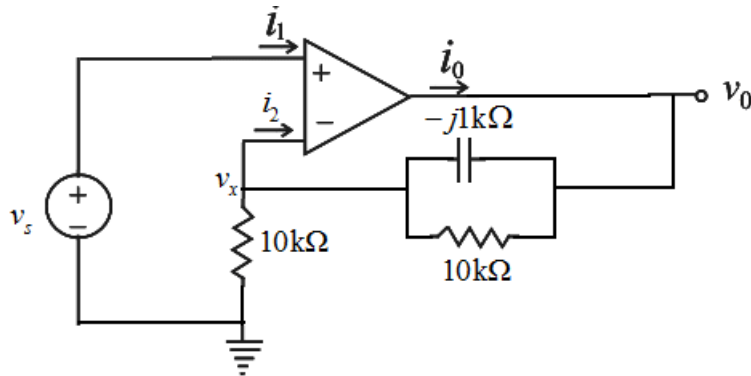
$$\omega = 1000 \text{ rad/sec}$$

$$1\text{ }\mu\text{F} \Rightarrow \frac{1}{j \times 1000 \times 1 \times 10^{-6}}$$

$$1\text{ }\mu\text{F} = -j1 \text{ k}\Omega$$

Step 2 of 4

Thus, the frequency domain equivalent circuit is,



From the above circuit,

Inverting non-inverting node voltages are equal i.e.,

$$v_x = v_s$$

Currents through inverting and non-inverting terminals are zero i.e.,

$$i_1 = i_2 = 0 \text{ A}$$

Step 3 of 4

Applying KCL at node v_x

$$\frac{v_x}{10 \times 10^3} + \frac{v_x - v_0}{Z_1} = 0 \quad \dots\dots\dots(1)$$

Where,

$$Z_1 = -j1 \text{ k}\Omega \parallel 10 \text{ k}\Omega$$

$$Z_1 = \frac{-j(1 \times 10^3) \times (10 \times 10^3)}{(10 \times 10^3) - j(1 \times 10^3)}$$

$$Z_1 = \frac{-j(10 \times 10^6)}{(10 - j1) \times 10^3}$$

$$Z_1 = \frac{-j(10 \times 10^3)}{10 - j1}$$

$$Z_1 = 0.099 - j0.99 \text{ k}\Omega$$

Step 4 of 4

From (1),

$$\frac{v_x}{10 \times 10^3} + \frac{v_x - v_0}{(0.099 - j0.99) \times 10^3} = 0$$

$$(0.099 - j0.99)v_x + 10v_x - 10v_0 = 0$$

$$(10.099 - j0.99)v_x = 10v_0 \quad (\because v_x = v_s)$$

$$\frac{v_s}{v_0} = \frac{10}{10.099 - j0.99}$$

$$\frac{1}{G} = \frac{v_s}{v_0} = 0.98078 + j0.0961$$

$$G = \frac{v_0}{v_s}$$

$$= \frac{1}{0.98078 + j0.0961}$$

$$G = 1.0099 - j0.099$$

$$\therefore G = 1.015 \angle -5.6^\circ$$

Thus, the closed – loop, gain is 1.015 and the phase shift is -5.6°